Business Problem Statement

The problem this bootcamp set out to solve

> Bootcamp-generated business case

See [data architecture and data flow diagrams](data_architecture.md).

Date: 2026-10-06 Project: test-claude-26 Design Pattern: Custom (bootcamp-generated scenario)

Problem Description

A retailer holds the same customers across several systems under different spellings, formats and levels of completeness, so it cannot get a trustworthy customer count or see one customer's full activity.

Use Case Category

Customer 360

Design Pattern Reference

None selected.

Data Sources

1. CRM_CONTACTS: Synthetic CRM export / ~4,000 records / People / Static snapshot generated for the bootcamp / Local file under data/raw/. The name is a single full-name field and the address is free text. 2. WEB_ORDERS: Synthetic e-commerce orders export / ~4,500 records / People / Static snapshot generated for the bootcamp / Local file under data/raw/. The name is split into parts and the address is split into street, city, state and postal code. 3. SUPPORT_TICKETS: Synthetic support-ticket export / ~1,500 records / People / Static snapshot generated for the bootcamp / Local file under data/raw/. Records are sparse (many with no address or phone) with misspellings and nicknames.

Entity Types

People

Key Matching Criteria

- Name (High priority): the main attribute shared by all three sources, but written differently in each.
- Email (High priority): a strong identifier where present.
- Phone (Medium priority): useful corroboration, but missing from many support-ticket records.
- Address (Medium priority): free text in one source and parts in another, so it needs mapping before it can be compared.

Quality concerns: missing and off-pattern values are spread across all three sources, and the sources do not share the same shape for names and addresses.

Success Criteria

- Report the number of resolved customers against the roughly 10,000 input records, so the duplication

across systems is visible.

- Show cross-source overlap: which customers appear in more than one source.
- Be able to explain why any two records were, or were not, resolved to the same customer.

Desired Output

Format: Master list, API, Reports Use case: One-time local load for the bootcamp Integration: Standalone

Integration Requirements

Downstream systems: none (integration_targets is empty in config/bootcamp_preferences.yaml)

Deployment Target

Platform: Local / on-premises Category: Local. Development proceeds locally first; infrastructure is a production follow-up covered by the graduation production project and migration checklist.

Timeline

Target completion: not specified Key milestones: the bootcamp modules

Notes

- The data is synthetic and generated for the bootcamp; no real customer data is involved.
- The scenario's roughly 10,000 records exceed the built-in evaluation capacity, so Data collection checks licensing before any load.